

RESEARCH ARTICLE | JANUARY 09 2026

Anisotropic diffusion of hydrogen dopants in crystalline InAlZnO

Gil Su Jeon  ; Jeong Wook Kim; Byoung Don Kong  *J. Appl. Phys.* 139, 023101 (2026)<https://doi.org/10.1063/5.0303334>

Articles You May Be Interested In

Effects of Al contents on the electrical performance and low-frequency noise properties of InAlZnO thin-film transistors

Appl. Phys. Lett. (October 2025)

Effect of dopants on grain boundary decohesion of Ni: A first-principles study

Appl. Phys. Lett. (December 2008)

A predictive modeling study of the impact of chemical doping on the strength of a Ag/ZnO interface

J. Appl. Phys. (December 2018)



Freedom to Innovate.

The New VHFLI 200 MHz Lock-in Amplifier.

Orchestrate pulses, triggers, and acquisition as the hub of your experiment. Discover more – run every signal analysis tool, simultaneously.

Order now

Anisotropic diffusion of hydrogen dopants in crystalline InAlZnO

Cite as: J. Appl. Phys. 139, 023101 (2026); doi: 10.1063/5.0303334

Submitted: 21 September 2025 · Accepted: 17 December 2025 ·

Published Online: 9 January 2026



Gil Su Jeon,¹ Jeong Wook Kim,¹ and Byoung Don Kong^{1,2,3,a)}

AFFILIATIONS

¹Department of Electrical Engineering, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea

²Department of Semiconductor Engineering, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea

³Graduate School of Semiconductor Technology, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea

^{a)}Author to whom correspondence should be addressed: bdkong@postech.ac.kr

ABSTRACT

Hydrogen incorporation strongly affects carrier density and reliability in oxide semiconductors, yet its behavior in the crystalline indium–aluminum–zinc oxide has not been fully established. Using first-principles calculations, stable hydrogen incorporation sites and their migration characteristics were systematically examined. Hydrogen preferentially occupies oxygen atoms coordinated by three zinc atoms, which remain stable against dissociation with barriers exceeding 0.5 eV. In contrast to the crystalline indium–gallium–zinc oxide, where gallium and zinc readily intermix, the crystalline indium–aluminum–zinc oxide contains a distinct planar aluminum–oxygen layer that acts as a vertical diffusion barrier. Penetration across this dense hexagonal layer requires hydroxyl rotation rather than conventional dissociation–migration, leading to substantially higher activation energies. The combination of site selectivity and anisotropic migration enables confinement of hydrogen dopants within a sub-unit cell thickness, providing a pathway to localized conductivity enhancement while maintaining the wide bandgap and low carrier density of the bulk material.

10 March 2026 14:35:27

© 2026 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0303334>

I. INTRODUCTION

Oxide semiconductors are widely studied as channel materials for displays,¹ power electronics,² and integrated circuits.³ Amorphous indium–gallium–zinc oxide (a-IGZO) has been widely adopted due to its high mobility and uniform electrical properties despite the absence of long-range order.^{4,5} Recent advances in atomic layer deposition have enabled the growth of crystalline oxide semiconductors with precise cation control.^{6–8} Among them, crystalline indium–aluminum–zinc oxide (c-IAZO) is of particular interest for memory and optical applications,^{9–11} where Al incorporation reduces vacancy-related leakage and widens the bandgap, enhancing retention and transparency via stronger Al–O bonds than Ga–O.¹²

Hydrogen is known to play a decisive role in oxide semiconductors, acting as a shallow donor or by passivating defects such as oxygen vacancies and dangling bonds.^{13–18} Hydrogenation can improve device reliability by reducing traps but may also increase

carrier density, leading to instability.^{19–22} These trade-offs highlight the need for spatially selective hydrogen control within transistors. This is especially relevant for c-IAZO, where unintended hydrogen incorporation from precursors or processing gases may diffuse deeply due to its low atomic mass, compromising the low off-current required for memory and display applications.^{23,24}

Prior works on c-IGZO treated Ga–O and Zn–O sublattices as intermixed layers,^{25,26} consistent with the chemical similarity of Ga and Zn and supported by theoretical interpretations of experimental observation that can be well explained by site disorder.^{27,28} In contrast, such disorder has not been reported for c-IAZO, likely due to the more pronounced differences in ionic radius and oxygen affinity between Al and Zn relative to Ga and Zn,¹² thereby stabilizing a distinct Al–O layer. This structural distinction implies that hydrogen behavior in c-IAZO may differ fundamentally from that in c-IGZO, requiring dedicated investigation.

In this work, the stable hydrogen incorporation sites in c-IAZO are identified, and the migration barriers between them are quantified. Particular emphasis is placed on anisotropic diffusion associated with the planar Al–O layer, which emerges as a vertical barrier that confines hydrogen dopants and governs their impact on electronic properties.

II. COMPUTATIONAL METHODS

All density functional theory (DFT) calculations were performed using the Quantum ESPRESSO package.²⁹ The Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation (GGA)³⁰ was employed for the exchange–correlation functional, providing reliable energy profiles at moderate computational cost without empirical parameter tuning, although it is known to underestimate the bandgaps.^{31–33} The plane wave basis was expanded up to 50 Ry with a charge-density cutoff of 500 Ry to generate a fine real-space grid spacing of 0.07 Å. This resolution, sufficiently finer than the Bohr radius (0.529 Å), was chosen to accurately describe the localized ionic states, particularly those involving hydrogen.

A. Structural and electronic calculations

Both pristine c-IAZO (In:Al:Zn:O = 1:1:1:4) primitive cell and the H-doped supercells were relaxed until residual atomic forces fell below 0.01 eV/Å. The Brillouin zone was sampled over a $15 \times 15 \times 6$ Monkhorst–Pack mesh³⁴ for the primitive c-IAZO cell and proportionally reduced for larger supercells to maintain equivalent k-point density. Band structure calculations were subsequently carried out on the relaxed geometries along the high-symmetry path of the hexagonal lattice, whereas the density of states (DOS) were obtained from additional non-self-consistent field calculations using a k-point grid twice as dense along each reciprocal axis to achieve accurate sampling of the Brillouin zone and a smooth DOS profile. The projected density of states (PDOS) were also derived to analyze the contributions from each atomic orbital. Details specific to each dopant configuration are described in the corresponding sections.

B. Ionic state analysis

The ionic states of atoms were evaluated using Bader charge analysis.³⁵ The charge density obtained by the self-consistent field calculation was divided into atomic basins bounded by zero-flux surfaces of the charge-density gradients, as implemented in the Henkelman charge-analysis code.³⁶ The deviation of the integrated electron count within each basin from the nominal valence electrons was taken as the ionic state of each atom.

C. Determination of minimum-energy pathways

The migration pathways of hydrogen and the corresponding activation energies (E_A) were investigated using the nudged elastic band (NEB) method.³⁷ Relaxed H-doped supercells served as the initial and final images, and a sufficient number of intermediate images were inserted to ensure that the trajectory of the path remained smooth and continuous with inter-image distance shorter than the Bohr radius.

Because the most efficient migration occurs when displacements follow the direction of the acting force, atomic positions in each intermediate image were iteratively updated along the NEB force direction. This optimization proceeded until the force components orthogonal to the path tangent were reduced below 0.01 eV/Å. The minimum energy path was then refined with the climbing-image (CI) technique,³⁸ which accurately locates the saddle point along the trajectory. E_A for forward and reverse migration was obtained as the energy difference between the CI and the respective initial or final state.

III. STRUCTURAL PROPERTIES AND STABLE H SITES IN c-IAZO

A. Crystal model of c-IAZO

The lattice structure of c-IAZO (In:Al:Zn:O = 1:1:1:4) was modeled based on the established cation-site preferences observed in homologous ternary oxides.³⁹ Larger trivalent cations (e.g., In^{3+}) preferentially occupy octahedral sites coordinated by six oxygen atoms, whereas smaller ones (e.g., Al^{3+} , Ga^{3+}) favor trigonal-bipyramidal sites with fivefold coordination. Divalent cations (e.g., Zn^{2+}) are accommodated at tetrahedral sites with fourfold coordination. Following these established trends, the initial c-IAZO structure was constructed with In, Al, and Zn, respectively, occupying octahedral, trigonal-bipyramidal, and tetrahedral sites, and relaxed.

The resulting c-IAZO structure is shown in Fig. 1, where the four oxygen sites (O1–O4) are labeled along the *c* axis: O1 between the In and Al layers, O2 within the Al layer, O3 between the Al and Zn layers, and O4 between the Zn and In layers.

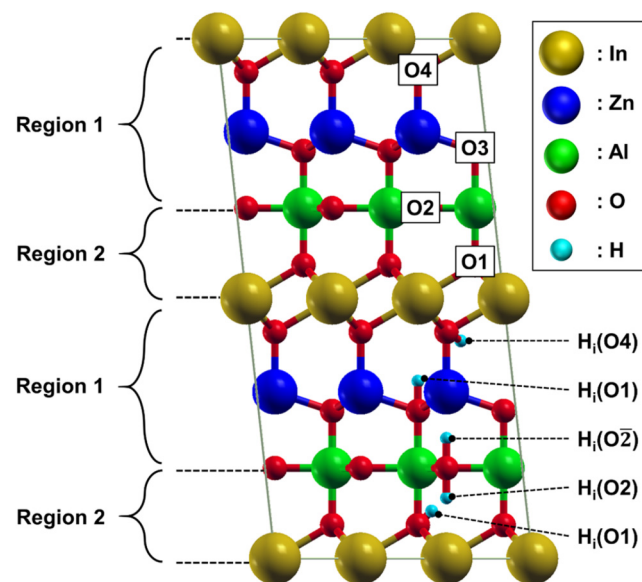


FIG. 1. Crystal structure of c-IAZO with distinct O atoms labeled according to their crystallographic environment. A H atom can either bind to each O site forming OH bonds, or substitute the O atom at a vacancy site.

10 March 2026 14:35:27

TABLE I. Coordination number and bond length between each metal and O atoms.

	n_{O1}	n_{O2}	n_{O3}	n_{O4}
In [d_{InO} (Å)]	3 (2.203)	0	0	3 (2.201)
Zn [d_{ZnO} (Å)]	0	0	3 (2.004)	1 (1.991)
Al [d_{AlO} (Å)]	1 (1.893)	3 (1.901)	1 (1.900)	0

The optimized atomic coordinates for undoped c-IAZO are summarized in Table I. In this table, the partial coordination number of an atom with respect to a surrounding atom type X (In, Al, Zn, or O1–O4) is denoted as n_X , and the bond length between atoms Y and Z is denoted as d_{YZ} . A notable feature is that O2 is coordinated exclusively with Al and does not bond with Zn, maintaining a relatively long separation of 2.535 Å. This behavior contrasts sharply with c-IGZO, where the Ga–O and Zn–O sublattices are intermixed, and both Ga and Zn are typically fivefold coordinated with oxygen.

Table II summarizes the Bader charge analysis of c-IAZO. Among the metal cations, Zn shows the lowest ionization, consistent with its stable 2+ valence state and relatively weak bonding to oxygen. In contrast, Al is nearly fully ionized due to its strong Al–O bonding, which makes the O2 site the most negatively charged among O1–O4 even though it is coordinated only to three Al atoms and has no Zn neighbors. This exclusive Al–O coordination of O2 further supports the tendency of c-IAZO to maintain separated Al–O and Zn–O sublattices, rather than a mixed environment.

B. Stable OH-bonded interstitial H sites

To model interstitial hydrogen in c-IAZO, we introduced one additional H atom into $3 \times 3 \times 2$ supercells and relaxed. When placed near an oxygen site, this atom bonds to O and forms a protonic OH group; such interstitial hydrogen is denoted H_i in the following. The addition of H in this form both preserves charge neutrality of the supercell and reflects the typical incorporation of hydrogen in oxide semiconductors. The OH groups bonded to O1–O4 are labeled $H_i(O1)$ – $H_i(O4)$ (Fig. 1), and their bond lengths (d_{OH}) and relative energies (E_{rel}) are summarized in Table III. For O2, two bonding orientations are possible; the less stable configuration (along +z) is referred to as $H_i(O2)$. For clarity, the unit cell is divided by the Al–O2 plane into two zones: region 1 (O3–Al–O4–In) and region 2 (In–O1–Al–O2).

TABLE II. Net charge of each atom in a c-IAZO unit cell.

	Net charge (e)
In	+1.874
Zn	+1.259
Al	+2.929
O1	−1.452
O2	−1.910
O3	−1.473
O4	−1.229

TABLE III. Hydroxyl bond length and relative energy of the c-IAZO supercell with each H_i dopant configuration.

	d_{OH} (Å)	E_{rel} (eV)
$H_i(O1)$	1.028	0.310
$H_i(O2)$	0.9838	0.102
$H_i(O2)$	0.9830	0.321
$H_i(O3)$	0.9809	0
$H_i(O4)$	0.988	0.056

The chosen doping level (one H in 18 unit cells, $6.8 \times 10^{20} \text{ cm}^{-3}$) corresponds to an H–H separation of ~ 10 Å, sufficient to minimize artificial interactions. Bader charge analysis shows that H atoms at all interstitial sites are positively charged (+0.59e to +0.63e), consistent with their role as electron donors. A stable H_i configuration requires both a strong OH bond and minimal Coulombic repulsion with nearby cations. By these criteria, $H_i(O3)$ is the most stable site, having the shortest d_{OH} and open space toward the In atom along +z. $H_i(O4)$ is the next most stable, with a slightly longer d_{OH} but oriented toward the least ionized Zn cation. Both $H_i(O3)$ and $H_i(O4)$ are located in region 1.

C. Stable oxygen vacancy substitutional H sites

Substitutional hydrogen in c-IAZO was evaluated by considering H occupying oxygen-vacancy sites, denoted as H_O . Calculations were performed on $3 \times 3 \times 2$ supercells with one O atom replaced by H. Configurations with H at the O1–O4 sites are labeled $H_O(O1)$ – $H_O(O4)$, and their bond lengths to neighboring metals (d_{MH}) and E_{rel} are summarized in Table IV. Among these, $H_O(O3)$ was found to be the most stable and is, therefore, taken as the energy reference (0 eV).

The stability of a given H_O site depends on both the formation energy of the underlying oxygen vacancy and the strength of the H–metal bond formed at that site. Notably, only Zn exhibits a shorter bond length with H than with the original O, indicating the formation of a relatively stable Zn–H bond. As a result, the O3 site, associated with the weakest Zn–O bond, provides the most favorable substitutional site.

Bader charge analysis further shows that substitutional H atoms are nearly fully negatively charged, behaving as H^- anions that replace O^{2-} . An oxygen vacancy nominally introduces two excess electrons, one of which is typically trapped in a deep-level state.⁴⁰ When H occupies the vacancy, it acts as a monovalent

TABLE IV. Bond length with the adjacent metal atom and relative energy of the c-IAZO supercell with each H_O dopant configuration [M = In for $H_O(O1)$ and $H_O(O4)$, Al for $H_O(O2)$, and Zn for $H_O(O3)$].

	d_{MH} (Å)	E_{rel} (eV)
$H_O(O1)$	2.140	0.375
$H_O(O2)$	2.042	1.111
$H_O(O3)$	1.896	0
$H_O(O4)$	2.385	0.314

10 March 2026 14:35:27

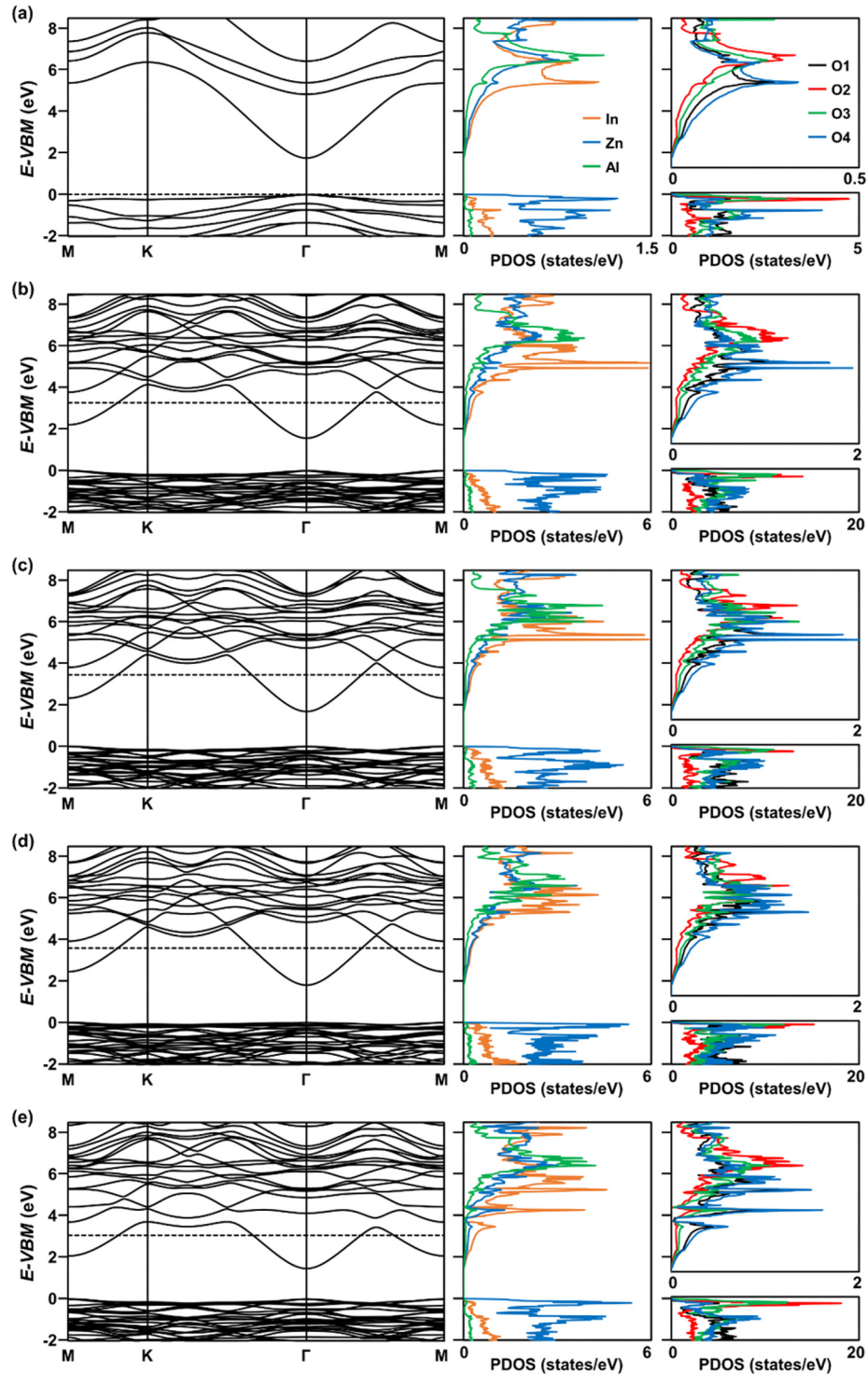


FIG. 2. Energy band structure and PDOS of c-lAZO in (a) the undoped form and with dopants as (b) H_i(O1), (c) H_i(O2), (d) H_i(O3), and (e) H_i(O4). The VBM of each structure is set to 0 eV, and the dotted lines indicate E_F . The metal contributions, O contributions for valence and conduction bands are shown separately to accommodate the large differences in their magnitudes. The contribution from H is omitted due to the negligible magnitude relative to other elements.

anion that passivates the deep state, while the remaining electron is released as a shallow donor.

IV. ELECTRONIC STRUCTURE OF H-DOPED c-IAZO

A. H_i configurations

The band structures and PDOS calculated for the identified stable H_i sites directly quantify the effects of interstitial doping on c-IAZO (Fig. 2). Owing to the well-known bandgap underestimation of the GGA-PBE functional, the calculated electronic structures are discussed only in a comparative manner, focusing on relative changes among doping configurations rather than on absolute bandgap values. As in a-IGZO, the conduction band minimum (CBM) is dominated by In 5s orbitals, while the valence band maximum (VBM) arises from O 2p states.⁴¹ Introducing H_i does not generate midgap defect states and preserves the direct Γ -point transition, but produces site-dependent modifications in the PDOS near the Fermi level (E_F) along with a modest bandgap reduction. These features mainly arise from the metal atoms adjacent to the dopant: In for $H_i(O1)$ and $H_i(O4)$, and Zn for $H_i(O3)$.

Key electronic parameters were extracted from the band structures and PDOS (Table V). The electron effective mass was obtained by fitting the parabolic dispersion near the CBM, and the fractional contributions of In, Zn, and Al to the conduction band were quantified by integrating their PDOS up to E_F (or up to CBM +1 eV for the undoped case where $E_F < \text{CBM}$).

The site dependence of H_i doping is particularly significant. When H bonds to O in the In–O layer (O1 or O4), the bandgap is markedly reduced, reflecting the central role of the In–O sublattice in forming the conduction band. In contrast, $H_i(O3)$ primarily interacts with Zn, leading to a reduced DOS near the CBM, a higher E_F , and a more dispersive conduction band. $H_i(O4)$, however, enhances the DOS near the CBM through strong In interaction, resulting in the narrowest bandgap, a lower E_F , and a flatter conduction band. Thus, the two most stable H_i sites, O3 and O4, induce nearly opposite effects on the electronic parameters despite their small energy difference. To quantify their macroscopic impact, a detailed evaluation of E_A for site-to-site transitions is, therefore, required.

B. H_O configurations

The electronic structures obtained for c-IAZO with H_O configurations reveal the influence of substitutional dopants on c-IAZO (Fig. 3). In all cases, H_O doping produces flatter valence

band dispersion near the VBM and a larger bandgap compared with the H_i -doped structures. This behavior originates from the perturbation of the O 2p-derived valence band by the H 1s orbital, which gives rise to a sharp H-derived PDOS peak near the VBM. The impact is most pronounced for the $H_O(O3)$ configuration, where the strong Zn–H bond leads to an inverted, downward-convex curvature near the Γ -point and the most intense H-related PDOS feature. In the conduction band region close to E_F , site-dependent contributions from neighboring metal atoms also become more prominent, including Al-derived states near $H_O(O2)$.

The corresponding electronic parameters are summarized in Table VI. Relative to undoped c-IAZO, H_O doping increases the bandgap further, partly due to the larger average interatomic distances induced by oxygen vacancies within the fixed lattice volume. The charge-density distributions confirm that electrons near the H_O site are more localized owing to the smaller atomic radius of H compared with O, leading to a notable increase in effective mass. Neighboring metal contributions to the conduction band are also enhanced, more strongly than in the H_i -doped cases. Among the substitutional sites, $H_O(O3)$ remains the most favorable configuration and exhibits the lowest effective mass, similar to the trend observed for $H_i(O3)$.

V. ACTIVATION ENERGIES OF H DIFFUSION IN c-IAZO

A. Migration of H dopants between H_i sites

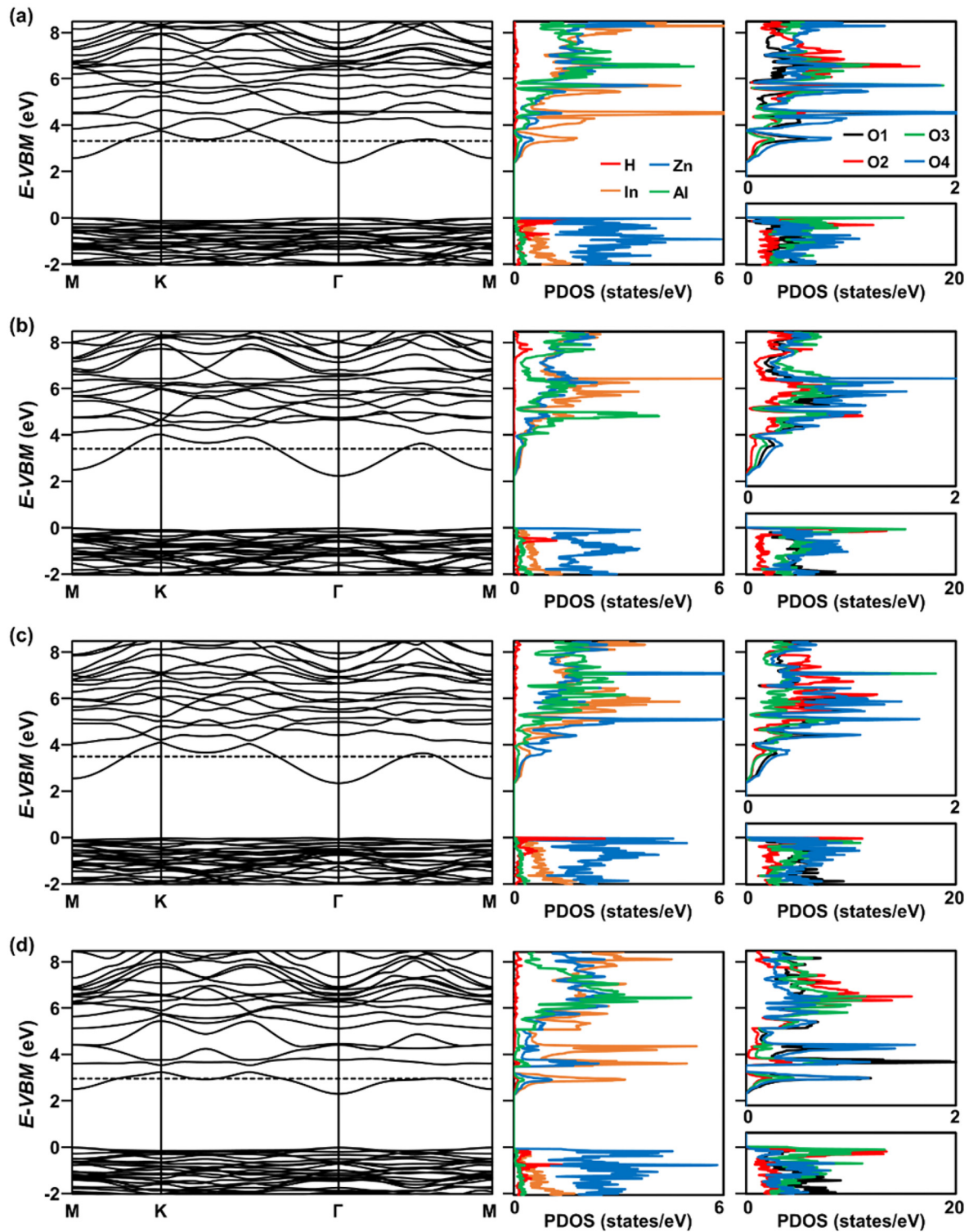
Intra-cell diffusion paths between O atoms within the same region were first determined using the NEB method (Fig. 4). The reliability of the calculated paths was verified by mapping the energy landscape along the $H_i(O1) \rightarrow H_i(O2)$ trajectory; the resulting contour map confirmed that the minimum energy path is orthogonal to iso-energy lines and that the CI saddle point is accurately located [Figs. 4(a) and 4(b)]. The $H_i(O1) \rightarrow H_i(O2)$ path exhibits a nearly barrier-less transition [Fig. 4(c)], whereas the $H_i(O3) \rightarrow H_i(O4)$ and $H_i(O3) \rightarrow H_i(O2)$ paths [Figs. 4(d) and 4(e)] show significantly higher activation barriers, exceeding 0.5 eV and slightly lower—values consistent with those reported for wurtzite ZnO ^{42,43} and a-IGZO.⁴⁴ This consistency, regardless of the presence or type of the trivalent cations, further reinforces the preference of H_i for the Zn-related O3 site and suggests that the lateral migration barrier primarily arises from breaking the stable OH bond rather than Coulombic repulsion from the neighboring cations. Because of the relatively high E_A of this barrier, once H stabilizes in the $H_i(O3)$ configuration during high-temperature annealing, it is unlikely to migrate into $H_i(O4)$ at room temperature. Consequently, $H_i(O3)$ can be regarded as the dominant interstitial site under equilibrium conditions, implying that the electronic structure of c-IAZO will remain largely unchanged without significant modifications to the bandgap or effective mass.

Macroscopic diffusion into neighboring unit cells was next investigated by analyzing paths from the most stable site in each region to the equivalent site in an adjacent cell (Fig. 5). In region 1, the calculated $H_i(O3) \rightarrow H_i(O2)$ barrier is lower than $H_i(O3) \rightarrow H_i(O4)$; however, because the $H_i(O2)$ state itself is energetically unstable, the saddle point is skewed toward $H_i(O2)$ and the effective diffusion path proceeds through the metastable $H_i(O4)$ configuration instead. When H is not bound to either O3 or O4, the system reaches an energy maximum, while $H_i(O4)$ serves as a

TABLE V. Comparison of electronic parameters of c-IAZO with the H_i doping configuration (m_e : rest mass of an electron).

	Bandgap (eV)	Effective mass (m_e)	Contribution ratio
			In:Zn:Al
Intrinsic	1.755	0.247	46:36:18
$H_i(O1)$	1.5732	0.243	52:33:15
$H_i(O2)$	1.6878	0.239	46:36:18
$H_i(O3)$	1.8161	0.242	45:38:17
$H_i(O4)$	1.4695	0.249	58:29:13

10 March 2026 14:35:27



10 March 2026 14:35:27

FIG. 3. Energy band structure and PDOS of c-IAZO with dopants as (a) $\text{H}_2\text{O}(\text{O1})$, (b) $\text{H}_2\text{O}(\text{O2})$, (c) $\text{H}_2\text{O}(\text{O3})$, and (d) $\text{H}_2\text{O}(\text{O4})$. The VBM of each structure is set to 0 eV, and the dotted lines indicate E_F . The contribution from H is also plotted as it produces a valence band peak in magnitude comparable with the metal elements.

TABLE VI. Comparison of electronic parameters of c-IAZO with the H₂O doping configuration (m_e : rest mass of an electron).

	Bandgap (eV)	Effective mass (m_e)	Contribution ratio In : Zn : Al
H ₂ O(O1)	2.3940	0.413	57:29:14
H ₂ O(O2)	2.2643	0.422	44:36:20
H ₂ O(O3)	2.3560	0.408	45:38:17
H ₂ O(O4)	2.3226	0.427	60:29:11

metastable state along the path. Region 2 shows a similar energy peak when H is not bonded to O1 or O2, but a distinct double-barrier profile is absent due to the instability of the H_i(O1) site. These results indicate that macroscopic diffusion is governed not simply by the lowest local barrier but by sequences of energetically stable configurations, with metastable states such as H_i(O4) dictating the effective migration pathway.

Vertical diffusion between region 1 and region 2 was also examined to evaluate migration across the Al–O plane (Fig. 6). The H_i(O4) → H_i(O1) path shows a high activation barrier despite the short O4–O1 distance, owing to significant distortion of the dense In₂O₃ layer as H penetrates through it. Similarly, the H_i(O2) → H_i(O2) rotational process yields a high barrier even

though the O–H bond length remains nearly unchanged. Bader charge analysis indicates that H retains its cationic state throughout this rotation; the barrier instead originates from breaking a strong Al–O bond and approaching a neighboring Al³⁺ cation, which produces strong Coulombic repulsion. Thus, inter-region diffusion is hindered because H must rotate within the OH group and cross the Al–O plane rather than follow a dissociation–migration pathway. This atomic-scale mechanism aligns with prior findings that a single atomic plane can function as a hydrogen barrier. In particular, single layer graphene—whose hexagonal lattice closely resembles the Al–O plane—exhibits low proton (H⁺) permeability and a high activation energy (~0.78 eV).⁴⁵ This analogy supports the role of the Al–O layer in c-IAZO as an intrinsic barrier against out-of-plane hydrogen transport.

To assess the influence of lattice variation, the H_i(O2) → H_i(O2) rotational path was recalculated with the lattice constant expanded up to 3.36 Å. As the lattice expands, Al–O bonds weaken and Coulombic repulsion between Al³⁺ and H⁺ decreases, leading to a substantial reduction in activation energy with only ~2% expansion [Fig. 6(e)]. The rotational barrier is higher in c-IGZO in the absence of structural disorder, which incorporates Ga with a larger ionic radius. In contrast, disordered c-IGZO, where Ga–O and Zn–O networks are intermixed, does not present a well-defined planar barrier and, therefore, allows lower-energy dissociation–migration pathways.^{46,47} These results indicate that the high vertical barrier in c-IAZO originates primarily from

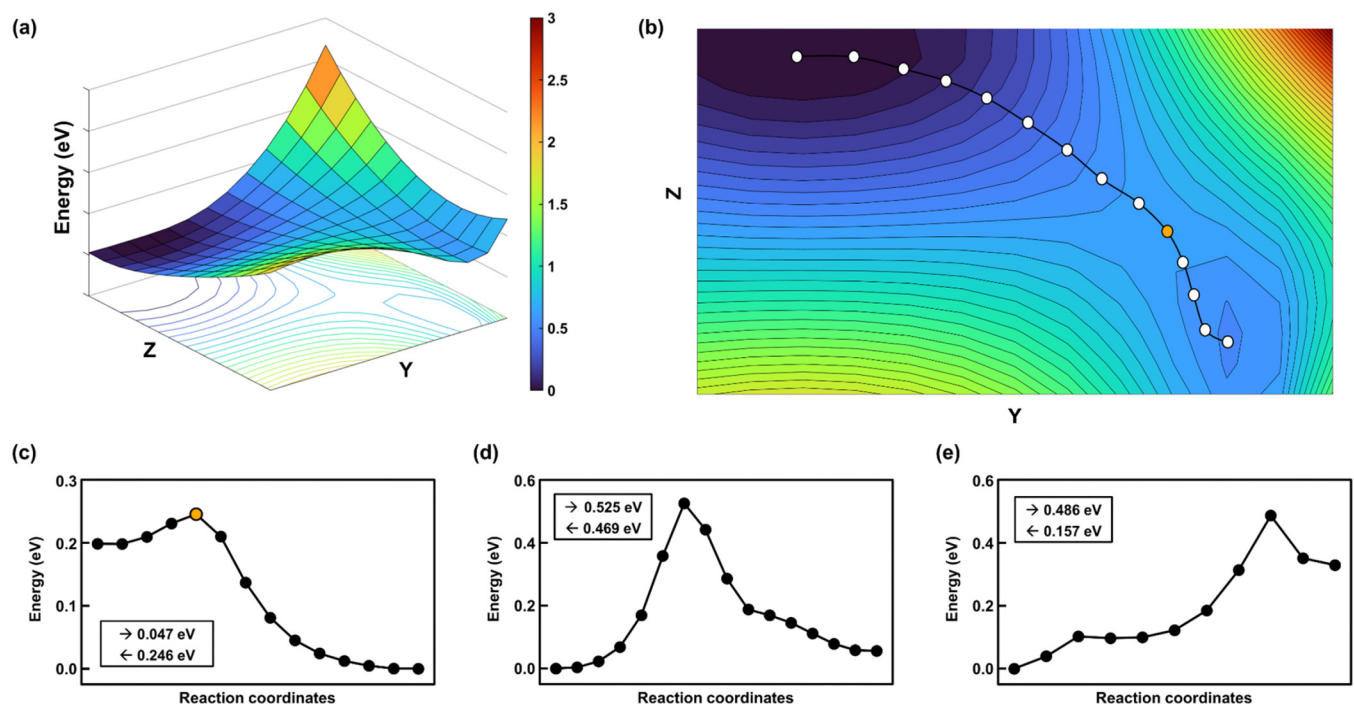
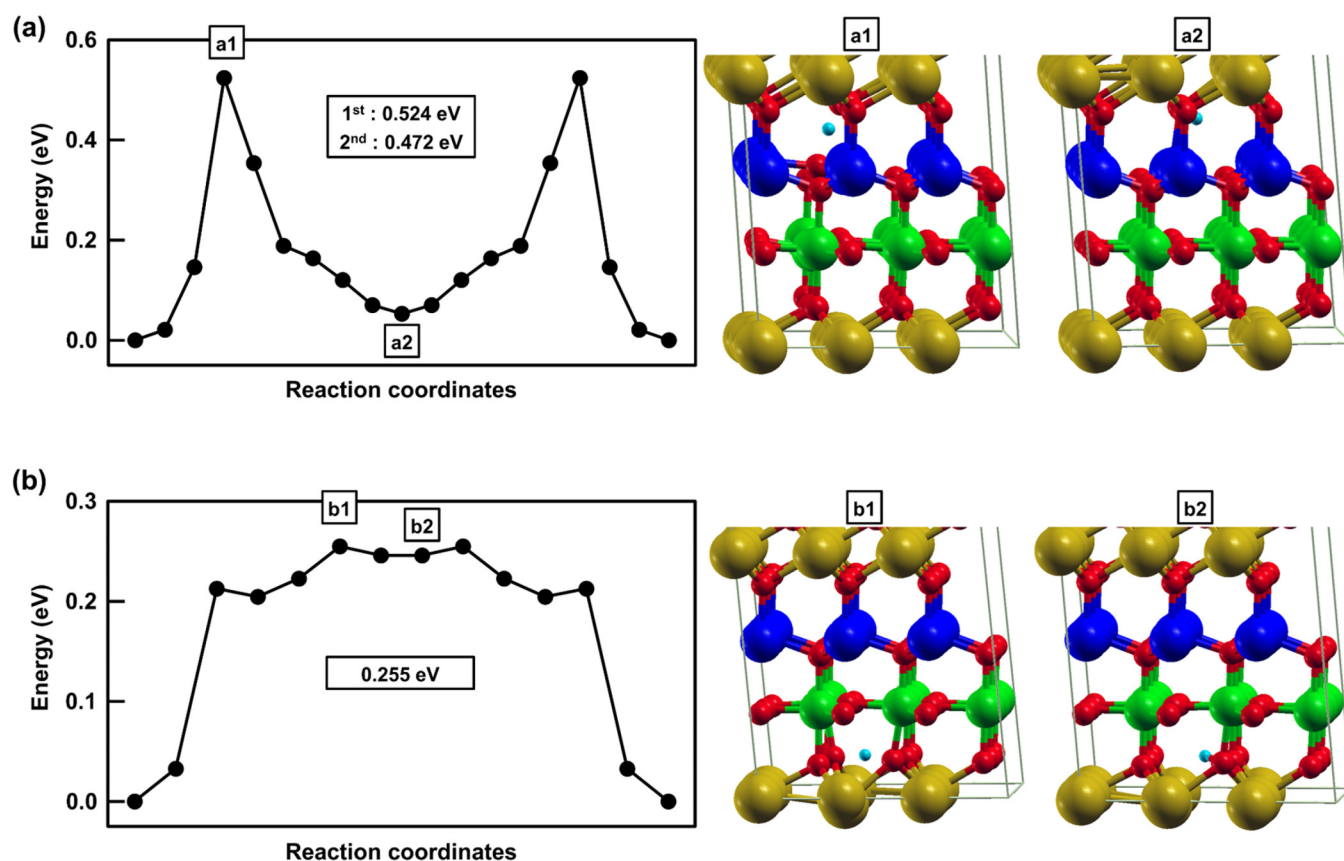


FIG. 4. (a) Crystal energy variation as a function of the H atom coordinates around O1 and O2. (b) Iso-energy contour map plotted with the diffusion path, where Cl at the saddle point is highlighted in yellow correspondingly as in (c). Energy profiles along H diffusion paths in c-IAZO: (c) H_i(O1) → H_i(O2), (d) H_i(O3) → H_i(O4), and (e) H_i(O3) → H_i(O2). E_a for the forward/reverse path is shown in the inset of each graph.

10 March 2026 14:35:27



10 March 2026 14:35:27

FIG. 5. Energy profile along inter-cell H diffusion paths with images of the crucial intermediate states: (a) $H_i(O3) \rightarrow H_i(O4) \rightarrow H_i(O3)$ and (b) $H_i(O2) \rightarrow H_i(O1) \rightarrow H_i(O2)$.

the ordered layer structure rather than elemental composition alone. Likewise, wurtzite ZnO, which cannot form such planar layers, has been previously reported to lack vertical anisotropy in hydrogen diffusion.^{42,43} This further highlights the significance of maintaining a dense and well-ordered Al-O plane in effectively suppressing inter-region H diffusion.

Overall, the migration of interstitial hydrogen in c-IAZO exhibits strong site- and region-dependence. The dominance of the $H_i(O3)$ site and the pronounced vertical anisotropy governed by the ordered Al-O layer collectively highlight that hydrogen transport in c-IAZO is controlled by both local bonding strength and lattice ordering. These findings establish the microscopic origin of hydrogen confinement and provide a foundation for comparison with H_O migration discussed in the following section.

B. Dissociation and migration of H dopants from H_O sites

E_A for H dissociating from oxygen-vacancy sites and migrating to nearby O atoms to form new O-H bonds were evaluated (Fig. 7). Among the substitutional configurations, $H_O(O3)$ is the

most stable, with a crystal energy more than 0.3 eV lower than other H_O sites, and can, therefore, be regarded as the dominant state under equilibrium. Migration pathways originating from $H_O(O3)$ were analyzed, including transitions to $H_i(O2)$ and $H_i(O4)$ within the same cell and to $H_i(O3)$ in a neighboring cell. Once the H atom leaves the vacancy and forms an OH bond, its subsequent diffusion follows the same mechanisms as the interstitial H_i paths described earlier.

All three calculated routes exhibit E_A significantly higher than those of H_i -to- H_i diffusion. In particular, the $H_O(O3) \rightarrow H_i(O4)$ transition exhibits a barrier comparable to values previously reported for wurtzite ZnO^{42,43} and c-IGZO,^{46,47} which can be attributed to the breaking of Zn-H bonds. Importantly, the $H_i(O3)$ and $H_i(O4)$ sites remain energetically favored even in the presence of vacancies, confirming that region 1 consistently acts as the preferred hydrogen incorporation region.

These results show that dissociation of H from vacancy sites is energetically costly, and once released, the dopant predominantly stabilizes in region 1 as $H_i(O3)$ or $H_i(O4)$. To quantify the impact of these migration barriers on macroscopic transport, the corresponding transition rates were next estimated using Arrhenius analysis.

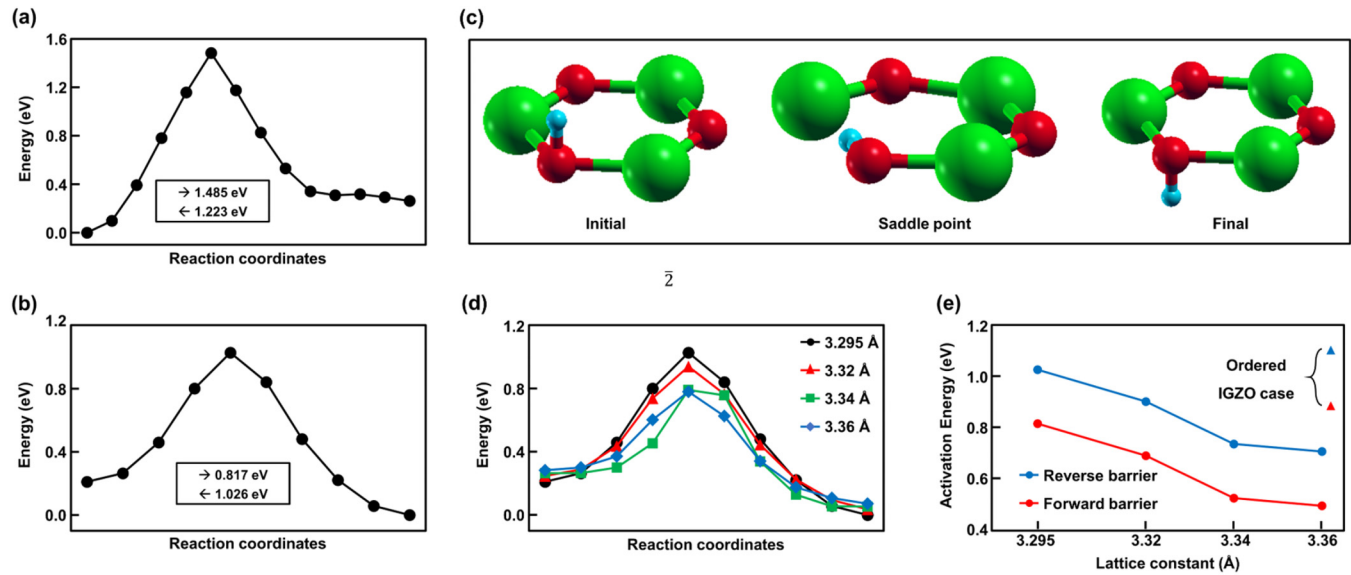


FIG. 6. Energy profile along inter-region hydrogen diffusion paths: (a) $H_i(O4) \rightarrow H_i(O1)$, and (b) $H_i(O2) \rightarrow H_i(O2)$. (c) Structure of the hexagonal Al-OH layer of the initial, intermediate and final state. Comparison of (d) energy profile along optimal pathways, and (e) forward/reverse E_A of the rotation process with increasing lattice constant from 3.295 to 3.36 Å. The forward/reverse E_A for c-IGZO without disorder is also plotted as triangles for comparison.

C. Transition rate estimation

The calculated E_A were further expressed in terms of transition rates to allow direct comparison of different migration processes. For a single-barrier path, the rate was estimated using the Arrhenius relation,

$$F = \nu \cdot \exp\left(-\frac{E_A}{k_B T}\right), \quad (1)$$

where k_B is the Boltzmann constant and a prefactor of $\nu = 10^{13} \text{ s}^{-1}$ was adopted as a typical attempt frequency for H diffusion in oxides, derived from the OH vibrational mode.⁴⁷ Although the precise prefactor may vary, particularly for rotational mechanism and dissociation from oxygen vacancy site, a uniform value was used here for practical comparison. For two-step processes, the

effective rate was evaluated as

$$F_{\text{series}} = \frac{F_1 \cdot F_2}{F_1 + F_2}. \quad (2)$$

Rates were quantified at three representative temperatures—300 K, 370 K (device operation), and 600 K (processing conditions)—and are summarized in Table VII.

These calculations highlight the highly asymmetric character of H migration in c-IAZO. Lateral diffusion within region 1 is more than three orders of magnitude slower than in region 2 below operating temperatures, suggesting that region 1 provides a more stable environment that may store H dopants effectively. Vertical diffusion is further suppressed by the additional barrier at the Al-O plane, with transition rates more than 10^4 times lower

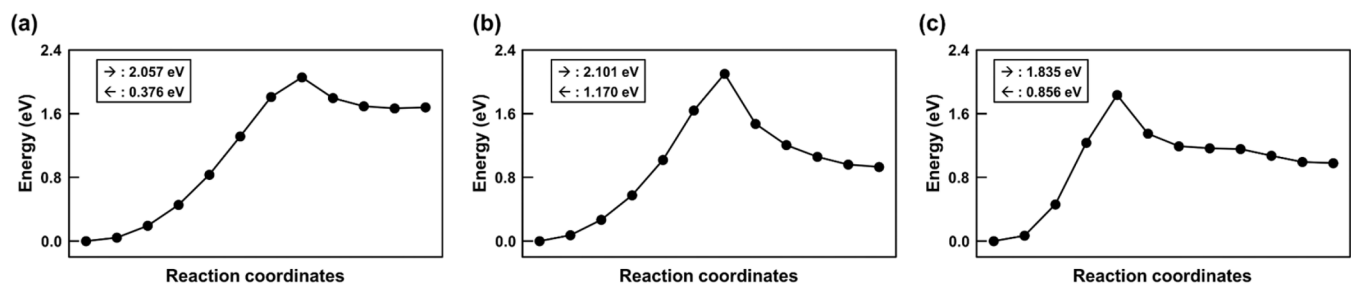


FIG. 7. Energy profile along the diffusion path from $H_O(O3)$ to adjacent O atoms, forming OH bonds: (a) $H_i(O2)$, (b) $H_i(O3)$, and (c) $H_i(O4)$.

TABLE VII. Reaction frequency of each diffusion path by temperature.

Diffusion pathway	F (s^{-1}) (300 K)	F (s^{-1}) (370 K)	F (s^{-1}) (600 K)
Lateral diffusion in region 1	1.34×10^4	5.94×10^5	2.86×10^8
Lateral diffusion in region 2	5.20×10^8	3.36×10^9	7.09×10^{10}
Vertical diffusion through the Al–O layer	1.87×10^{-1}	7.43×10^1	1.36×10^6
Vertical diffusion through the In–O layer	1.13×10^{-12}	5.93×10^{-8}	3.36×10^0
Dissociation from the passivated $H_O(O3)$ site	1.49×10^{-18}	1.01×10^{-12}	3.86×10^{-3}

than lateral diffusion at room temperature. This barrier effectively confines H dopants to region 1, where the $H_i(O3)$ and $H_i(O4)$ sites are most stable. However, because the barrier across the Al–O plane remains lower than that associated with the In–O network or with H_O dissociation, it may still serve as the most vulnerable pathway for vertical leakage. Its effectiveness as a confinement layer, therefore, depends on maintaining a dense, mismatch-free Al–O plane during deposition, and a more precise determination of the prefactor ν for rotational processes will be necessary for quantitative predictions.

The results also suggest practical implications for device design. Under typical doping levels, region 1 provides a stable environment that can accommodate H without significantly altering the electronic structure. At very high H concentrations, however—such as in heavily doped source/drain regions aimed at reducing contact resistance—the lateral migration barrier may not be sufficient to fully suppress migration into the channel. Moreover, less-stable $H_i(O4)$ states with lower dissociation energies may form as the H concentration increases and approaches the number of unit cells, enhancing diffusion. Deliberately creating oxygen vacancies in the source/drain regions could then serve as a strategy to localize H dopants and mitigate unwanted channel diffusion, particularly in short-channel devices.

VI. CONCLUSION

The stability and migration behavior of hydrogen in c-IAZO were clarified through first-principles calculations. Both interstitial H_i (OH-bonded) and substitutional H_O (at O vacancies) preferentially occupy the O3 site, coordinated mainly by Zn atoms, making this the dominant configuration under equilibrium conditions. Once formed, O3-related sites exhibit comparatively high migration barriers, indicating that they are not only energetically stable but also resistant to dissociation. This points to the Zn–O sublattice as the primary host for H incorporation in c-IAZO.

The layered structure of c-IAZO further introduces strong anisotropy in hydrogen migration. The ordered Al–O plane acts as an effective vertical diffusion barrier, suppressing inter-region transport by more than four orders of magnitude relative to lateral diffusion. This barrier is highly sensitive to lattice expansion and structural disorder, emphasizing the importance of preserving a dense and well-matched Al–O sublattice during growth.

Together, these results highlight the critical role of site-specific stability and anisotropic migration in determining the distribution of H dopants in c-IAZO. The insights provide not only a microscopic understanding of H behavior in oxide semiconductors but

also practical guidance for confining H in desired regions and enhancing device stability.

ACKNOWLEDGMENTS

This work was partly supported by the Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (No. RS-2024-00401466, HRD Program for Industrial Innovation) and National Research Foundation (NRF) grant funded by the Korea Government (MIST) (No. RS-2025-16069611).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Gil Su Jeon: Conceptualization (supporting); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (supporting). **Jeong Wook Kim:** Methodology (supporting); Software (supporting). **Byoung Don Kong:** Conceptualization (lead); Data curation (supporting); Formal analysis (equal); Funding acquisition (lead); Investigation (supporting); Methodology (supporting); Project administration (lead); Resources (lead); Supervision (lead); Validation (equal); Writing – original draft (supporting); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- 1E. Fortunato, P. Barquinha, and R. Martins, *Adv. Mater.* **24**(22), 2945–2986 (2012).
- 2M. Higashiwaki, K. Sasaki, A. Kuramata, T. Masui, and S. Yamakoshi, *Phys. Status Solidi A* **211**(1), 21–26 (2014).
- 3S. Datta, S. Dutta, B. Grisafe, J. Smith, S. Srinivasa, and H. Ye, *IEEE Micro* **39**(6), 8–15 (2019).
- 4K. Nomura, H. Ohta, A. Takagi, T. Kamiya, M. Hirano, and H. Hosono, *Nature* **432**(7016), 488–492 (2004).
- 5T. Kamiya and H. Hosono, *NPG Asia Mater.* **2**(1), 15–22 (2010).

10 March 2026 14:35:27

- ⁶M. H. Cho, H. Seol, A. Song, S. Choi, Y. Song, P. S. Yun, K. B. Chung, J. U. Bae, K. S. Park, and J. K. Jeong, *IEEE Trans. Electron Device* **66**(4), 1783–1788 (2019).
- ⁷Y. S. Kim, H. W. Kim, T. Hwang, J. Ahn, S. B. Cho, and J. S. Park, *Adv. Electron. Mater.* **11**(15), e00137 (2025).
- ⁸J. Xie, J. Shi, M. Zhang, J. Dong, D. Han, and X. Zhang, *Appl. Surf. Sci.* **682**, 161744 (2025).
- ⁹H. Fujiwara, Y. Sato, N. Saito, T. Ueda, and K. Ikeda, *IEEE Trans. Electron Devices* **67**(12), 5329–5335 (2020).
- ¹⁰W. Xiong, B. Luo, W. Meng, B. Zhu, X. Wu, and S. J. Ding, *Sci. China Inform. Sci.* **67**(9), 199401 (2024).
- ¹¹L. Yang, B. Luo, X. Chen, W. Xiong, M. Yang, W. Meng, J. Teng, B. Zhu, S. J. Ding, and X. Wu, *IEEE Trans. Electron Devices* **72**(5), 2305–2311 (2025).
- ¹²R. D. Shannon, *Found. Crystallogr.* **32**(5), 751–767 (1976).
- ¹³K. Nomura, T. Kamiya, and H. Hosono, *ECS J. Solid State Sci. Technol.* **2**(1), P5 (2013).
- ¹⁴A. Song, H. W. Park, K. B. Chung, Y. S. Rim, K. S. Son, J. H. Lim, and H. Y. Chu, *Appl. Phys. Lett.* **111**(24), 243507 (2017).
- ¹⁵Y. Hanyu, K. Domen, K. Nomura, H. Hiramatsu, H. Kumomi, H. Hosono, and T. Kamiya, *Appl. Phys. Lett.* **103**(20), 202114 (2013).
- ¹⁶J. Kim, S. Bang, S. Lee, S. Shin, J. Park, H. Seo, and H. Jeon, *J. Mater. Res.* **27**(17), 2318–2325 (2012).
- ¹⁷S. I. Oh, G. Choi, H. Hwang, W. Lu, and J. H. Jang, *IEEE Trans. Electron Devices* **60**(8), 2537–2541 (2013).
- ¹⁸A. Sato, K. Abe, R. Hayashi, H. Kumomi, K. Nomura, T. Kamiya, M. Hirano, and H. Hosono, *Appl. Phys. Lett.* **94**(13), 133502 (2009).
- ¹⁹B. Du Ahn, H. S. Shin, H. J. Kim, J. S. Park, and J. K. Jeong, *Appl. Phys. Lett.* **93**(20), 203506 (2008).
- ²⁰S. W. Tsao, T. C. Chang, S. Y. Huang, M. C. Chen, S. C. Chen, C. T. Tsai, Y. J. Kuo, Y. C. Chen, and W. C. Wu, *Solid-State Electron.* **54**(12), 1497–1499 (2010).
- ²¹T. Miyase, K. Watanabe, I. Sakaguchi, N. Ohashi, K. Domen, K. Nomura, H. Hiramatsu, H. Kumomi, H. Hosono, and T. Kamiya, *ECS J. Solid State Sci. Technol.* **3**(9), Q3085 (2014).
- ²²A. Abliz, *J. Alloys Compd.* **831**, 154694 (2020).
- ²³H. Y. Kim, S. I. Cho, D. Y. Shin, K. B. Chung, S. H. K. Park, and J. B. Ko, *J. Alloys Compd.* **1020**, 179353 (2025).
- ²⁴P. Y. Yen, K. J. Zhou, P. J. Sun, Y. A. Chen, Y. T. Chien, B. S. Huang, P.-Y. Lee, T.-H. Juan, K. You Shao, Y. Shan Lin, S. M. Sze, and T.-C. Chang, *IEEE Electron Device Lett.* **45**(9), 1586–1589 (2024).
- ²⁵A. Walsh, J. L. F. Da Silva, and S. H. Wei, *Chem. Mater.* **21**(21), 5119–5124 (2009).
- ²⁶C. Chen, K. C. Cheng, E. Chagarov, and J. Kanicki, *Jpn. J. Appl. Phys.* **50**(9), 091102 (2011).
- ²⁷Y. Waseda, K. Sugiyama, and T. Kawamata, *Mater. Trans.* **59**(11), 1691–1700 (2018).
- ²⁸Y. Jegal, A. H. Kulahlioglu, C. K. Baek, and B. D. Kong, *IEEE Electron Device Lett.* **41**(6), 872–875 (2020).
- ²⁹P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, A. Dal Corso, S. de Gironcoli, S. Fabris, G. Fratesi, R. Gebauer, U. Gerstmann, C. Gougoussis, A. Kokalj, M. Lazzeri, L. Martin-Samos, N. Marzari, F. Mauri, R. Mazzarello, S. Paolini, A. Pasquarello, L. Paulatto, C. Sbraccia, S. Scandolo, G. Sclauzero, A. P. Seitsonen, A. Smogunov, P. Umari, and R. M. Wentzcovitch, *J. Phys.: Condens. Matter* **21**(39), 395502 (2009).
- ³⁰J. P. Perdew, K. Burke, and M. Ernzerhof, *Phys. Rev. Lett.* **77**(18), 3865 (1996).
- ³¹M. Liu, A. Gopakumar, V. I. Hegde, J. He, and C. Wolverton, *Phys. Rev. Mater.* **8**(4), 043803 (2024).
- ³²R. Devi, B. Singh, P. Canepa, and G. Sai Gautam, *npj Comput. Mater.* **8**(1), 160 (2022).
- ³³P. Deák, M. Lorke, B. Aradi, and T. Frauenheim, *J. Appl. Phys.* **126**(13), 130901 (2019).
- ³⁴H. J. Monkhorst and J. D. Pack, *Phys. Rev. B* **13**(12), 5188 (1976).
- ³⁵R. F. W. Bader, *Chem. Rev.* **91**(5), 893–928 (1991).
- ³⁶G. Henkelman, A. Arnaldsson, and H. Jónsson, *Comput. Mater. Sci.* **36**(3), 354–360 (2006).
- ³⁷G. Henkelman and H. Jónsson, *J. Chem. Phys.* **113**(22), 9978–9985 (2000).
- ³⁸G. Henkelman, H. Jónsson, B. P. Uberuaga, and H. Jónsson, *J. Chem. Phys.* **113**(22), 9901–9904 (2000).
- ³⁹J. L. F. Da Silva, Y. Yan, and S. H. Wei, *Phys. Rev. Lett.* **100**(25), 255501 (2008).
- ⁴⁰T. Kamiya, K. Nomura, and H. Hosono, *Phys. Status Solidi A* **207**(7), 1698–1703 (2010).
- ⁴¹T. Kamiya, K. Nomura, M. Hirano, and H. Hosono, *Phys. Status Solidi C* **5**(9), 3098–3100 (2008).
- ⁴²M. G. Wardle, J. P. Goss, and P. R. Briddon, *Phys. Rev. Lett.* **96**(20), 205504 (2006).
- ⁴³J. Bang and K. J. Chang, *Appl. Phys. Lett.* **92**(13), 132109 (2008).
- ⁴⁴H. K. Noh, J. S. Park, and K. J. Chang, *J. Appl. Phys.* **113**(6), 063712 (2013).
- ⁴⁵S. Hu, M. Lozada-Hidalgo, F. C. Wang, A. Mishchenko, F. Schedin, R. R. Nair, E. W. Hill, D. W. Boukhvalov, M. I. Katsnelson, R. A. W. Dryfe, I. V. Grigorieva, H. A. Wu, and A. K. Geim, *Nature* **516**(7530), 227–230 (2014).
- ⁴⁶T. Kanagawa, M. Takahashi, T. Nakayama, and S. Yamazaki, in *The 2019 International Conference on Solid State Devices and Materials* (The Japan Society of Applied Physics, 2019), pp. 309–310.
- ⁴⁷M. Nakashima, M. Oota, N. Ishihara, Y. Nonaka, T. Hirohashi, M. Takahashi, S. Yamazaki, T. Obonai, Y. Hosaka, and J. Koezuka, *J. Appl. Phys.* **116**(21), 213703 (2014).